MODULE 06 SUMMARY

Geometric Optics

Big Picture

Many of the phenomena we perceive involve light and matter. Scientists explore how light reflects, refracts, propagates and interact with matter, and these images is more and more important.

The course begins with understanding reflection of mirrors, moving on to refraction, dispersion, total internal reflection, and finishing with the study of lenses. Students will learn mathematical reasoning, hypothesis, and experimental methodology as used in modern science, and how to apply these skills to real-world problems.

Essential Standard

Students will explore the nature of light, including its reflection, refraction, and propagation, and how light interacts with matter. Students will use mirrors, lenses, and other optical instruments, and they will develop critical thinking and the ability to ask questions.

Top 10 Core Ideas

Concept	Description
Law of Reflection	$\theta_i = \theta_r$
Law of Refraction (Snell's Law)	$n_1 \sin\theta_1 = n_2 \sin\theta_2$
Critical Angle	$\sin\theta_c = \frac{n_2}{n_1}$
Mirror Equation	$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$
Magnification	$m = -\frac{d_i}{d_o} = \frac{h_i}{h_o}$
Lensmaker's Equation	$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$
Lens Equation	$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$
Focal Length	$f = \frac{R}{2}$
Total Internal Reflection	Occurs when light travels from higher to lower refractive index and $\theta_i > \theta_c$
Snell's Law	$n_1 \sin\theta_1 = n_2 \sin\theta_2$
Mirror Equation	$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$
Magnification	$m = -\frac{d_i}{d_o} = \frac{h_i}{h_o}$

Section 6.1 — Reflection and the Law of Reflection

Key Concepts

- Light travels in straight rays
- Reflection occurs at a boundary
- The angle of incidence equals the angle of reflection
- Reflection can be specular (smooth) or diffuse (rough)

Vocabulary

- Incident Ray
- Reflected Ray
- Normal
- Angle of Incidence
- Angle of Reflection
- Specular Reflection
- Diffuse Reflection

Formula

Law of Reflection

$$\theta_i = \theta_r$$

Representative Questions

1. **Calculation**
   Given the angle of incidence, find the angle of reflection.
2. **Diagram**
   Draw the incident ray, reflected ray, and normal.
3. **Conceptual**
   Why do we see the moon? Explain how the normal relates to the Earth's surface?
4. **Conceptual**
   Why do we see our own reflection in a smooth spoon but not in a wooden spoon?

Vocabulary

- Incident Ray
- Reflected Ray
- Normal
- Angle of Incidence
- Angle of Reflection
- Specular Reflection
- Diffuse Reflection

Formula

Law of Reflection

$$\theta_i = \theta_r$$

Representative Questions

1. **Calculation**
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3. **Conceptual**
   Why do we see the moon? Explain how the normal relates to the Earth's surface?
4. **Conceptual**
   Why do we see our own reflection in a smooth spoon but not in a wooden spoon?

Section 6.2 — Refraction and Snell's Law

Key Concepts

- Light changes speed when it enters a new medium
- Light changes direction when it enters a new medium
- Snell's Law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$
- Angle of incidence
- Angle of refraction
- Angle of incidence equals the angle of refraction
- Light bends toward the normal when it slows down, bends away from the normal when it speeds up

Vocabulary

- Refraction
- Refractive Index
- Incident Ray
- Refracted Ray
- Normal
- Angle of Incidence
- Angle of Refraction

Formula

Refraction Index

$$n = \frac{c}{v}$$

Snell's Law

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Representative Questions

1. **Calculation**
   Given the angle of incidence, find the angle of refraction.
2. **Diagram**
   Draw the incident ray, refracted ray, and normal.
3. **Conceptual**
   Why does light change direction when it enters a new medium?
4. **Conceptual**
   Why does light bend toward the normal when it slows down?

Vocabulary

- Refraction
- Refractive Index
- Incident Ray
- Refracted Ray
- Normal
- Angle of Incidence
- Angle of Refraction

Formula

Refraction Index

$$n = \frac{c}{v}$$

Snell's Law

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Representative Questions

1. **Calculation**
   Given the angle of incidence, find the angle of refraction.
2. **Diagram**
   Draw the incident ray, refracted ray, and normal.
3. **Conceptual**
   Why does light change direction when it enters a new medium?
4. **Conceptual**
   Why does light bend toward the normal when it slows down?

Section 6.3 — Total Internal Reflection and the Critical Angle

Key Concepts

- Total internal reflection occurs when light travels from a higher refractive index to a lower refractive index
- The critical angle is the minimum angle of incidence that produces a refracted angle of 90°
- The angle of incidence must be greater than the critical angle
- Total internal reflection is used in fiber optics and medical endoscopes

Vocabulary

- Critical Angle
- Total Internal Reflection
- Refractive Index

Formula

Critical Angle

$$\sin\theta_c = \frac{n_2}{n_1}$$

Critical Angle Condition

$$\sin\theta_i > \frac{n_2}{n_1}$$

Representative Questions

1. **Calculation**
   Given the refractive index, find the critical angle.
2. **Conceptual**
   Why does total internal reflection only occur when light travels from a higher refractive index to a lower refractive index?
3. **Conceptual**
   Why is the critical angle the minimum angle of incidence that produces a refracted angle of 90°?
4. **Conceptual**
   Why must the angle of incidence be greater than the critical angle for total internal reflection to occur?

Vocabulary

- Critical Angle
- Total Internal Reflection
- Refractive Index

Formula

Critical Angle

$$\sin\theta_c = \frac{n_2}{n_1}$$

Critical Angle Condition

$$\sin\theta_i > \frac{n_2}{n_1}$$

Representative Questions

1. **Calculation**
   Given the refractive index, find the critical angle.
2. **Conceptual**
   Why does total internal reflection only occur when light travels from a higher refractive index to a lower refractive index?
3. **Conceptual**
   Why is the critical angle the minimum angle of incidence that produces a refracted angle of 90°?
4. **Conceptual**
   Why must the angle of incidence be greater than the critical angle for total internal reflection to occur?

Section 6.4 — Mirrors: Plane, Concave, and Convex

Key Concepts

- Plane mirrors produce virtual, upright images of the same size as the object
- Concave mirrors can produce real or virtual images depending on the object distance
- Convex mirrors always produce virtual, upright, and diminished images
- The mirror equation relates object distance, image distance, and focal length
- Magnification determines image size and orientation
- Sign conventions are critical: real images have positive d_i, virtual images have negative d_i

Vocabulary

- Plane Mirror
- Concave Mirror
- Convex Mirror
- Focal Point
- Focal Length
- Real Image
- Virtual Image

Formula

Mirror Equation

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

Magnification

$$m = -\frac{d_i}{d_o} = \frac{h_i}{h_o}$$

Focal Length

$$f = \frac{R}{2}$$

Representative Questions

1. **Calculation**
   Given the focal length and object distance, find the image distance.
2. **Conceptual**
   Why do concave mirrors always produce real images?
3. **Conceptual**
   Why do convex mirrors always produce virtual images?
4. **Conceptual**
   Why is the focal length of a convex mirror considered negative?

Vocabulary

- Plane Mirror
- Concave Mirror
- Convex Mirror
- Focal Point
- Focal Length
- Real Image
- Virtual Image

Formula

Mirror Equation

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

Magnification

$$m = -\frac{d_i}{d_o} = \frac{h_i}{h_o}$$

Focal Length

$$f = \frac{R}{2}$$

Representative Questions

1. **Calculation**
   Given the focal length and object distance, find the image distance.
2. **Conceptual**
   Why do concave mirrors always produce real images?
3. **Conceptual**
   Why do convex mirrors always produce virtual images?
4. **Conceptual**
   Why is the focal length of a convex mirror considered negative?

Section 6.5 — Lenses: Double Convex and Double Concave

Key Concepts

- Converging (double convex) lenses bring parallel rays to a focal point
- Diverging (double concave) lenses spread parallel rays apart
- The lensmaker's equation relates focal length to lens properties
- The lens equation relates object distance, image distance, and focal length
- Magnification determines image size and orientation
- Sign conventions are critical: real images have positive d_i, virtual images have negative d_i

Vocabulary

- Double Convex Lens
- Double Concave Lens
- Converging Lens
- Diverging Lens
- Focal Length
- Focal Point
- Real Image
- Virtual Image

Formula

Lens Equation

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

Magnification

$$m = -\frac{d_i}{d_o} = \frac{h_i}{h_o}$$

Representative Questions

1. **Calculation**
   Given the focal length and object distance, find the image distance.
2. **Conceptual**
   Why do converging lenses always produce real images?
3. **Conceptual**
   Why do diverging lenses always produce virtual images?
4. **Conceptual**
   Why do converging lenses have positive focal length and diverging lenses have negative focal length?

Vocabulary

- Double Convex Lens
- Double Concave Lens
- Converging Lens
- Diverging Lens
- Focal Length
- Focal Point
- Real Image
- Virtual Image

Formula

Lens Equation

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

Magnification

$$m = -\frac{d_i}{d_o} = \frac{h_i}{h_o}$$

Representative Questions

1. **Calculation**
   Given the focal length and object distance, find the image distance.
2. **Conceptual**
   Why do converging lenses always produce real images?
3. **Conceptual**
   Why do diverging lenses always produce virtual images?
4. **Conceptual**
   Why do converging lenses have positive focal length and diverging lenses have negative focal length?

Cumulative Mixed Review — Version 1

1. **Reflection Law**
   Given a ray at 40° to the normal, what is the angle of reflection?
2. **Critical Angle**
   Light goes from glass ($n = 1.5$) to air. Find the critical angle.
3. **Mirror Equation**
   $f = 10$ cm, $d_o = 30$ cm. Find d_i.
4. **Lensmaker's Equation**
   $n = 1.5$, $R_1 = 20$ cm, $R_2 = -20$ cm. Find f.
5. **Snell's Law**
   Air ($n = 1.00$) to water ($n = 1.33$), $\theta_i = 30°$. Find the angle of refraction.
6. **Magnification**
   $d_o = 20$ cm, $d_i = -10$ cm, $h_o = 5$ cm. Find m and h_i.
7. **Lens Equation**
   $f = 15$ cm, $d_o = 45$ cm. Find d_i.
8. **Total Internal Reflection**
   Glass ($n = 1.5$) to air. Is 45° greater than the critical angle?
9. **Image Characteristics**
   For $d_o = 40$ cm, $f = 10$ cm, describe the image: real or virtual, upright or inverted, magnified or diminished.
10. **Focal Length**
    $R = 40$ cm. Find f.

Cumulative Mixed Review — Version 2

1. **Reflection Law**
   Given a ray at 60° to the normal, what is the angle of reflection?
2. **Critical Angle**
   Light goes from diamond ($n = 2.4$) to air. Find the critical angle.
3. **Mirror Equation**
   $f = 15$ cm, $d_o = 45$ cm. Find d_i.
4. **Lensmaker's Equation**
   $n = 1.5$, $R_1 = 30$ cm, $R_2 = -30$ cm. Find f.
5. **Snell's Law**
   Air ($n = 1.00$) to glass ($n = 1.5$), $\theta_i = 45°$. Find the angle of refraction.
6. **Magnification**
   $d_o = 30$ cm, $d_i = -15$ cm, $h_o = 4$ cm. Find m and h_i.
7. **Lens Equation**
   $f = 20$ cm, $d_o = 60$ cm. Find d_i.
8. **Total Internal Reflection**
   Glass ($n = 1.5$) to air. Is 60° greater than the critical angle?
9. **Image Characteristics**
   For $d_o = 60$ cm, $f = 20$ cm, describe the image: real or virtual, upright or inverted, magnified or diminished.
10. **Focal Length**
    $R = 60$ cm. Find f.

Cumulative Mixed Review — Version 3

1. **Reflection Law**
   Given a ray at 30° to the normal, what is the angle of reflection?
2. **Critical Angle**
   Light goes from water ($n = 1.33$) to air. Find the critical angle.
3. **Mirror Equation**
   $f = 20$ cm, $d_o = 60$ cm. Find d_i.
4. **Lensmaker's Equation**
   $n = 1.5$, $R_1 = 40$ cm, $R_2 = -40$ cm. Find f.
5. **Snell's Law**
   Air ($n = 1.00$) to glass ($n = 1.5$), $\theta_i = 60°$. Find the angle of refraction.
6. **Magnification**
   $d_o = 40$ cm, $d_i = -20$ cm, $h_o = 6$ cm. Find m and h_i.
7. **Lens Equation**
   $f = 25$ cm, $d_o = 75$ cm. Find d_i.
8. **Total Internal Reflection**
   Glass ($n = 1.5$) to air. Is 30° greater than the critical angle?
9. **Image Characteristics**
   For $d_o = 80$ cm, $f = 20$ cm, describe the image: real or virtual, upright or inverted, magnified or diminished.
10. **Focal Length**
    $R = 80$ cm. Find f.

Cumulative Mixed Review — Answer Key

Version 1 — Solutions

1. The angle of incidence equals the angle of reflection. Both are measured relative to the normal.
2. $\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) = \sin^{-1}\left(\frac{1.0}{1.5}\right) = 41.8°$
3. Rearrange the mirror equation: $\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{10} - \frac{1}{30} = \frac{2}{30}$, so $d_i = 15$ cm. Positive means real image.
4. Lensmaker's Equation: $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (1.5-1)\left(\frac{1}{20} - \frac{1}{-20}\right) = 0.5 \times \frac{2}{20} = \frac{1}{20}$, so $f = 20$ cm.
5. Snell's Law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$: $1.00 \sin 30° = 1.33 \sin\theta_2$, $\sin\theta_2 = \frac{0.50}{1.33} = 0.376$, so $\theta_2 = 22.1°$.
6. Mirror equation: $\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{-10} - \frac{1}{20} = -\frac{3}{20}$, so $d_i = -6.7$ cm. Magnification: $m = -\frac{d_i}{d_o} = -\frac{-6.7}{20} = 0.33$, so $h_i = m h_o = 0.33 \times 5 = 1.7$ cm.
7. Lens Equation: $\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{15} - \frac{1}{45} = \frac{2}{45}$, so $d_i = 22.5$ cm.
8. Glass ($n = 1.5$) to air. Is 45° greater than the critical angle? $\sin\theta_c = \frac{1.0}{1.5} = 0.667$, so $\theta_c = 41.8°$. Yes, 45° > 41.8°, so total internal reflection occurs.
9. $d_o = 40$ cm, $f = 10$ cm: $\frac{1}{d_i} = \frac{1}{10} - \frac{1}{40} = \frac{3}{40}$, so $d_i = 13.3$ cm. Positive, so real image. Since $d_i > d_o$, the image is magnified. Since real, it is inverted.
10. $R = 40$ cm. Find f. $f = \frac{R}{2} = \frac{40}{2} = 20$ cm.

Version 2 — Solutions

1. The angle of incidence equals the angle of reflection. Both are measured relative to the normal.
2. $\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) = \sin^{-1}\left(\frac{1.0}{2.4}\right) = 24.6°$
3. Rearrange the mirror equation: $\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{15} - \frac{1}{45} = \frac{2}{45}$, so $d_i = 22.5$ cm. Positive means real image.
4. Lensmaker's Equation: $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (1.5-1)\left(\frac{1}{30} - \frac{1}{-30}\right) = 0.5 \times \frac{2}{30} = \frac{1}{30}$, so $f = 30$ cm.
5. Snell's Law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$: $1.00 \sin 45° = 1.5 \sin\theta_2$, $\sin\theta_2 = \frac{0.707}{1.5} = 0.471$, so $\theta_2 = 28.1°$.
6. Mirror equation: $\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{-15} - \frac{1}{30} = -\frac{3}{30}$, so $d_i = -10$ cm. Magnification: $m = -\frac{d_i}{d_o} = -\frac{-10}{30} = 0.33$, so $h_i = m h_o = 0.33 \times 4 = 1.3$ cm.
7. Lens Equation: $\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{20} - \frac{1}{60} = \frac{2}{60}$, so $d_i = 30$ cm.
8. Glass ($n = 1.5$) to air. Is 60° greater than the critical angle? $\sin\theta_c = \frac{1.0}{1.5} = 0.667$, so $\theta_c = 41.8°$. Yes, 60° > 41.8°, so total internal reflection occurs.
9. $d_o = 60$ cm, $f = 20$ cm: $\frac{1}{d_i} = \frac{1}{20} - \frac{1}{60} = \frac{2}{60}$, so $d_i = 30$ cm. Positive, so real image. Since $d_i > d_o$, the image is magnified. Since real, it is inverted.
10. $R = 60$ cm. Find f. $f = \frac{R}{2} = \frac{60}{2} = 30$ cm.

Version 3 — Solutions

1. The angle of incidence equals the angle of reflection. Both are measured relative to the normal.
2. $\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) = \sin^{-1}\left(\frac{1.0}{1.33}\right) = 48.6°$
3. Rearrange the mirror equation: $\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{20} - \frac{1}{60} = \frac{2}{60}$, so $d_i = 30$ cm. Positive means real image.
4. Lensmaker's Equation: $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (1.5-1)\left(\frac{1}{40} - \frac{1}{-40}\right) = 0.5 \times \frac{2}{40} = \frac{1}{40}$, so $f = 40$ cm.
5. Snell's Law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$: $1.00 \sin 60° = 1.5 \sin\theta_2$, $\sin\theta_2 = \frac{0.866}{1.5} = 0.577$, so $\theta_2 = 35.3°$.
6. Mirror equation: $\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{-20} - \frac{1}{40} = -\frac{3}{40}$, so $d_i = -13.3$ cm. Magnification: $m = -\frac{d_i}{d_o} = -\frac{-13.3}{40} = 0.33$, so $h_i = m h_o = 0.33 \times 6 = 2.0$ cm.
7. Lens Equation: $\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{25} - \frac{1}{75} = \frac{2}{75}$, so $d_i = 37.5$ cm.
8. Glass ($n = 1.5$) to air. Is 30° greater than the critical angle? $\sin\theta_c = \frac{1.0}{1.5} = 0.667$, so $\theta_c = 41.8°$. No, 30° < 41.8°, so total internal reflection does not occur.
9. $d_o = 80$ cm, $f = 20$ cm: $\frac{1}{d_i} = \frac{1}{20} - \frac{1}{80} = \frac{3}{80}$, so $d_i = 26.7$ cm. Positive, so real image. Since $d_i > d_o$, the image is magnified. Since real, it is inverted.
10. $R = 80$ cm. Find f. $f = \frac{R}{2} = \frac{80}{2} = 40$ cm.